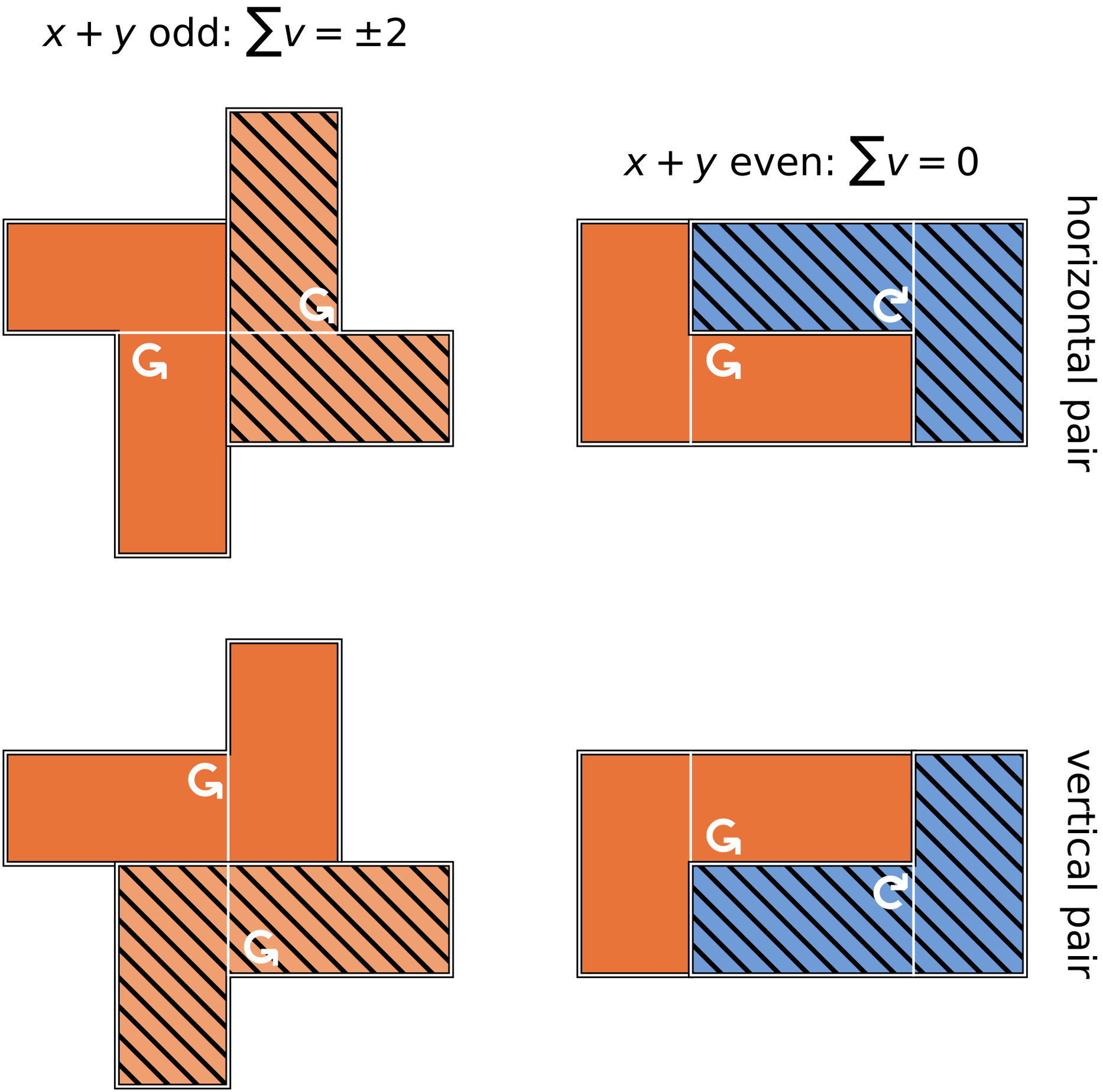
$x + y$ odd: $\sum v = \pm 2$
$x + y$ even: $\sum v = 0$
G
G
G
G
G
G
horizontal pair
vertical pair